

Complex Numbers $\rightarrow$ Matrix.

(Fibonacci Seq.)

□ Q.

Let
$$f(z) = \frac{z-1}{z+1}$$

Solve this equation

$$f^{(2026)}(z) = z$$

$$\underbrace{f(f(f \dots (z)))}_{2026 \text{ times}} = z$$

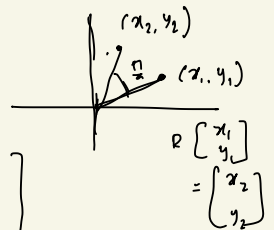
Theory:

$$z \rightarrow \frac{az+b}{cz+d} \quad (\text{Mobius eqn})$$

$$\downarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$z \xrightarrow{M} \frac{z-1}{z+1}$$

$$M = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$



$$\begin{matrix} \frac{\pi}{4} \\ \nwarrow \\ \frac{\pi}{4} \\ \swarrow \\ \frac{\pi}{4} \end{matrix}$$

$$= \sqrt{2} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \sqrt{2} \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix}$$

Rotation Matrix

$\rightarrow R$

$$M^{2026} = (\sqrt{2})^{2026} \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix}^{2026}$$

$$= 2^{1013} \begin{bmatrix} \cos (2026 \frac{\pi}{4}) & -\sin (2026 \frac{\pi}{4}) \\ \sin (2026 \frac{\pi}{4}) & \cos (2026 \frac{\pi}{4}) \end{bmatrix}$$

$$= 2^{2013} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

M^{2026}

$$= \begin{bmatrix} 0 & -2^{2013} \\ 2^{2013} & 0 \end{bmatrix}$$

$$f^{(2026)}(z) = \frac{0 \cdot z - 2^{2013}}{2^{2013} z + 0}$$

$$= -\frac{1}{z}$$

$$f^{(2026)}(z) = z \Rightarrow -\frac{1}{z} = z$$

$$\Rightarrow z^2 = -1 \Rightarrow z = \pm i$$

□ Determine the real value of c such that the function $f(z) = \frac{z+2}{cz+1}$ is of period 4.

$$[f^{(4)}(z) = z, \text{ } \forall \text{ all valid } z]$$

$$\text{but } \underline{f^{(2)}(z) \neq z}$$

$$\Rightarrow M = \begin{bmatrix} 1 & 2 \\ c & 1 \end{bmatrix}$$

$$\frac{z+2}{cz+1} \xrightarrow{M^4} z = \frac{z}{1} = \frac{kz+0}{0+k}$$

$$k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(kI)$$

(k - scalar)

$$\text{but } \underline{M^2 \neq cI}$$

$$X = M^2 = \begin{bmatrix} 1 & 2 \\ c & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ c & 1 \end{bmatrix} = \begin{bmatrix} 1+2c & 4 \\ 2c & 1+2c \end{bmatrix}$$

$$X^2 = (M^2)^2 = kI$$

char
eqn

$$\begin{aligned} & \text{tr}(X) = 0 \Rightarrow X^2 = kI \\ & \text{tr}(M^2) = 0 \Rightarrow c = -\frac{1}{2} \end{aligned}$$

$$M^2 = \begin{bmatrix} 0 & 4 \\ -1 & 0 \end{bmatrix}$$

$$\begin{aligned} \text{check: } M^4 &= \begin{bmatrix} 0 & 4 \\ -1 & 0 \end{bmatrix}^2 = \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 4 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix} = -4I. \end{aligned}$$

one ex:

$$f(z) = \frac{4z-2}{z+1}$$

$$\left(\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \right)^n = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix}$$

Find a closed form of $f^n(z)$

$$M = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$$

↓ evaluate
e-vector

$$(2, 3)$$

eigenvector comp $(\lambda = 2)$

$$M \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 2 \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2v_1 \\ 2v_2 \end{bmatrix}$$

$$\text{one such choice} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{e-value e-vector pairs: } (2, \begin{bmatrix} 1 \\ 1 \end{bmatrix}), (3, \begin{bmatrix} 2 \\ 1 \end{bmatrix})$$

(M^n) Diagonal Matrix

$$M = P D P^{-1}$$

$$M^n = P D^n P^{-1}$$

↑ Eigen decomposition.

$$M^2 = P D P^{-1} \overset{\lambda^2}{P D P^{-1}}$$

$$\downarrow = P D^2 P^{-1}$$

$$M^n = P D^n P^{-1}$$

$$P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$$

Then,

$$M = P D P^{-1}$$

$$\begin{aligned} \Rightarrow M^n &= P D^n P^{-1} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & 3^n \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2^n & 2 \cdot 3^n \\ 2^n & 3^n \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} -2^n + 2 \cdot 3^n & 2^{n+1} - 2 \cdot 3^n \\ -2^n + 3^n & 2^{n+1} - 3^n \end{bmatrix} \end{aligned}$$

$$\text{Hence } f^n(z) = \frac{(2 \cdot 3^n - 2^n)z + (2^{n+1} - 2 \cdot 3^n)}{(3^n - 2^n)z + (2^{n+1} - 3^n)}.$$

□ Q_n:

Let

$$f(z) = \frac{z-1}{z+1}$$

Solve:

$$f^{(5)}(z) = z$$

↓

$$f(z) = z \Rightarrow z^2 = -1 \Rightarrow z = \pm i$$

□

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$\xrightarrow{Q^n}$

$$\tilde{Q}^n = \begin{bmatrix} f_{n+1} & f_n \\ f_n & f_{n-1} \end{bmatrix}$$

f_n : n^{th}
fibonacci
number

Binomial Leap-Frogging :

$$\sum_{k=0}^n \binom{n}{k} 2^k f_k = f_{3n}$$

Pf:

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\Rightarrow \underline{Q^2} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow Q^3 = Q(Q+I) = \underline{Q^2 + Q} = 2Q + I$$

$$\begin{aligned} \Rightarrow Q^{3n} &= (\underline{2Q + I})^n \\ &= \sum_{k=0}^n \binom{n}{k} (2Q)^k \\ &= \sum_{k=0}^n \binom{n}{k} 2^k \underline{Q^k} \end{aligned}$$

$$\Rightarrow \begin{bmatrix} f_{3n+1} & \boxed{f_{3n}} \\ f_{3n} & f_{3n-1} \end{bmatrix} = \sum_{k=0}^n \binom{n}{k} 2^k \begin{pmatrix} f_{k+1} & f_k \\ f_k & f_{k-1} \end{pmatrix}$$

$$= \begin{pmatrix} \sum \binom{n}{k} 2^k f_{k+1} & \boxed{\sum \binom{n}{k} 2^k f_k} \\ \sum \binom{n}{k} 2^k f_k & \sum \binom{n}{k} 2^k f_{k-1} \end{pmatrix}$$

So, $\sum_{k=0}^n \binom{n}{k} 2^k \underline{f_k} = f_{3n}$

□

$$\boxed{f_{2n}}$$

$$= \sum_{k=0}^n \binom{n}{k} 3^{k-2} 2^{n-k} \underbrace{f_1}_{\boxed{f_4}} \underbrace{f_2}_{\boxed{f_3}} f_k$$

(gen)

$$f_{mn} = \sum_{k=0}^n \binom{n}{k} \underbrace{f_m^k f_{m-1}^{n-k}}_{\text{}} f_k$$

(m=2)

$$\begin{aligned} f_{2n} &= \sum_{k=0}^n \binom{n}{k} f_2^k f_1^{n-k} f_k \\ &= \sum_{k=0}^n \binom{n}{k} f_k \end{aligned}$$

□

$$f_n = \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f_{2k} \quad \leftarrow \text{Reverse Leap}$$

Derivation

$$Q^2 = Q + I$$

$$\Rightarrow Q = Q^2 - I$$

$$\Rightarrow Q^n = (Q^2 - I)^n$$

$$= \sum_{k=0}^n \binom{n}{k} Q^{2k} (-I)^{n-k}$$

$$= \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} Q^{2k}$$

$$\Rightarrow \begin{bmatrix} f_{n+1} & f_n \\ f_n & f_{n-1} \end{bmatrix} = \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} \begin{pmatrix} f_{2k+1} & f_{2k} \\ f_{2k} & f_{2k-1} \end{pmatrix}$$

$$\Rightarrow f_n = \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f_{2k}$$

□

$$f_n = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f_{3k}$$

Check:

$$Q^3 = 2Q + I$$

$$\Rightarrow Q = \frac{1}{2} (Q^3 - I)$$

$$\Rightarrow Q^n = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} Q^{3k}$$

$$\Rightarrow f_n = \frac{1}{2^n} \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f_{3k}$$

$$\boxed{f_n = \sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f_{mk}} \quad \begin{array}{c} \uparrow \\ \text{Find it (H.W.)} \end{array}$$